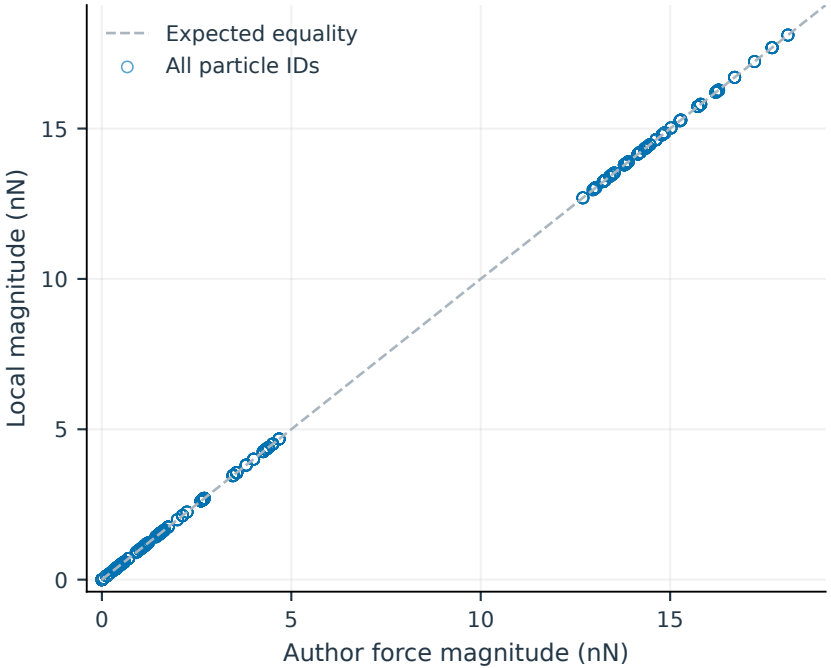
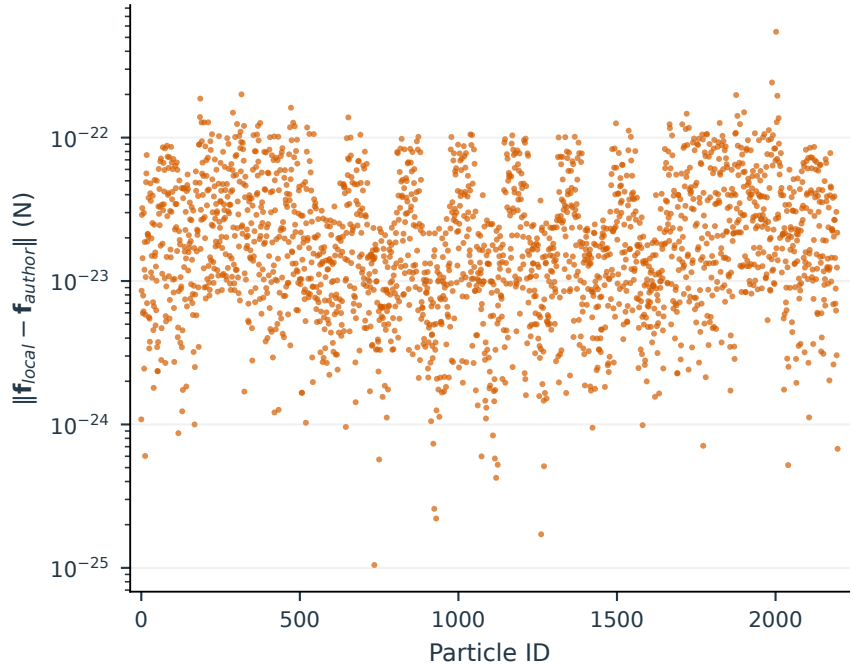
Force on every particle compared

2,197 aligned particle IDs | 6,591 force components | Existing local and author outputs

(a) Force-magnitude agreement



(b) Full-vector difference by particle



$$\|\mathbf{f}_{local} - \mathbf{f}_{author}\|_2 / \|\mathbf{f}_{local}\|_2 = \mathbf{5.861e\text{-}15}$$

Maximum per-particle absolute difference: 5.471e-22 N

Full-family core (27 particles): vector relative difference 2.322e-14

All particles retained. Differences use complete 3D vectors; no per-particle division by a near-zero force.